1 Cumulant Expression

$$\begin{aligned}\lambda^{pqr}_{stu} = \gamma^{pqr}_{stu} - \gamma^{p}_{s}\gamma^{qr}_{tu} + \gamma^{p}_{t}\gamma^{qr}_{su} + \gamma^{p}_{u}\gamma^{qr}_{ts} + \gamma^{q}_{s}\gamma^{pr}_{tu} - \gamma^{q}_{t}\gamma^{pr}_{su} + \gamma^{q}_{u}\gamma^{pr}_{st} + \gamma^{r}_{s}\gamma^{qp}_{tu} + \gamma^{r}_{t}\gamma^{pq}_{su} - \gamma^{r}_{u}\gamma^{pq}_{st} \\ + 2\gamma^{p}_{s}\gamma^{q}_{t}\gamma^{r}_{u} - 2\gamma^{p}_{s}\gamma^{q}_{u}\gamma^{r}_{t} - 2\gamma^{p}_{t}\gamma^{q}_{s}\gamma^{r}_{u} + 2\gamma^{p}_{t}\gamma^{q}_{u}\gamma^{r}_{s} + 2\gamma^{p}_{u}\gamma^{q}_{s}\gamma^{r}_{t} - 2\gamma^{p}_{u}\gamma^{q}_{t}\gamma^{r}_{s}\end{aligned} \tag{1}$$

2 Normal Order

$$\begin{aligned}a^{pqr}_{stu} = \gamma^{p}_{s}\tilde{a}^{qr}_{tu} - \gamma^{p}_{t}\tilde{a}^{qr}_{su} - \gamma^{p}_{u}\tilde{a}^{qr}_{ts} - \gamma^{q}_{s}\tilde{a}^{pr}_{tu} + \gamma^{q}_{t}\tilde{a}^{pr}_{su} \\ - \gamma^{q}_{u}\tilde{a}^{pr}_{st} - \gamma^{r}_{s}\tilde{a}^{qp}_{tu} - \gamma^{r}_{t}\tilde{a}^{pq}_{su} + \gamma^{r}_{u}\tilde{a}^{pq}_{st} + \gamma^{pq}_{st}\tilde{a}^{r}_{u} \\ - \gamma^{pq}_{su}\tilde{a}^{r}_{t} - \gamma^{pq}_{ut}\tilde{a}^{r}_{s} - \gamma^{pr}_{st}\tilde{a}^{q}_{u} + \gamma^{pr}_{su}\tilde{a}^{q}_{t} - \gamma^{pr}_{tu}\tilde{a}^{q}_{s} \\ - \gamma^{qr}_{su}\tilde{a}^{p}_{t} + \gamma^{qr}_{tu}\tilde{a}^{p}_{s} - \gamma^{rq}_{st}\tilde{a}^{p}_{u} + \gamma^{pqr}_{stu} + \tilde{a}^{pqr}_{stu}\end{aligned} \tag{2}$$

$$a^{pq}_{st} = \gamma^{p}_{s}\tilde{a}^{q}_{t} - \gamma^{p}_{t}\tilde{a}^{q}_{s} - \gamma^{q}_{s}\tilde{a}^{p}_{t} + \gamma^{q}_{t}\tilde{a}^{p}_{s} + \gamma^{pq}_{st} + \tilde{a}^{pq}_{st} \tag{3}$$

$$a^{p}_{s} = \gamma^{p}_{s} + \tilde{a}^{p}_{s} \tag{4}$$

3 Change to Operator

$$\tilde{a}^{p}_{s} = -\gamma^{p}_{s} + a^{p}_{s} \tag{5}$$

$$\tilde{a}^{pq}_{st} = -\gamma^{p}_{s}\tilde{a}^{q}_{t} + \gamma^{p}_{t}\tilde{a}^{q}_{s} + \gamma^{q}_{s}\tilde{a}^{p}_{t} - \gamma^{q}_{t}\tilde{a}^{p}_{s} - \gamma^{pq}_{st} + a^{pq}_{st} \tag{6}$$

4 Final Result

$$\begin{aligned}a^{pqr}_{stu} = 6\gamma^{p}_{s}\gamma^{q}_{t}\gamma^{r}_{u} - 2\gamma^{p}_{s}\gamma^{q}_{t}a^{r}_{u} - 6\gamma^{p}_{s}\gamma^{q}_{u}\gamma^{r}_{t} + 2\gamma^{p}_{s}\gamma^{q}_{u}a^{r}_{t} + 2\gamma^{p}_{s}\gamma^{r}_{t}a^{q}_{u} \\ - 2\gamma^{p}_{s}\gamma^{r}_{u}a^{q}_{t} - 2\gamma^{p}_{s}\gamma^{qr}_{tu} + \gamma^{p}_{s}a^{qr}_{tu} - 6\gamma^{p}_{t}\gamma^{q}_{s}\gamma^{r}_{u} + 2\gamma^{p}_{t}\gamma^{q}_{s}a^{r}_{u} \\ + 6\gamma^{p}_{t}\gamma^{q}_{u}\gamma^{r}_{s} - 2\gamma^{p}_{t}\gamma^{q}_{u}a^{r}_{s} - 2\gamma^{p}_{t}\gamma^{r}_{s}a^{q}_{u} + 2\gamma^{p}_{t}\gamma^{r}_{u}a^{q}_{s} + 2\gamma^{p}_{t}\gamma^{qr}_{su} \\ - \gamma^{p}_{t}a^{qr}_{su} + 6\gamma^{p}_{u}\gamma^{q}_{s}\gamma^{r}_{t} - 2\gamma^{p}_{u}\gamma^{q}_{s}a^{r}_{t} - 6\gamma^{p}_{u}\gamma^{q}_{t}\gamma^{r}_{s} + 2\gamma^{p}_{u}\gamma^{q}_{t}a^{r}_{s} \\ + 2\gamma^{p}_{u}\gamma^{r}_{s}a^{q}_{t} - 2\gamma^{p}_{u}\gamma^{r}_{t}a^{q}_{s} + \gamma^{p}_{u}\gamma^{qr}_{ts} + \gamma^{p}_{u}\gamma^{rq}_{st} - \gamma^{p}_{u}a^{qr}_{ts} \\ - 2\gamma^{q}_{s}\gamma^{r}_{t}a^{p}_{u} + 2\gamma^{q}_{s}\gamma^{r}_{u}a^{p}_{t} + 2\gamma^{q}_{s}\gamma^{pr}_{tu} - \gamma^{q}_{s}a^{pr}_{tu} + 2\gamma^{q}_{t}\gamma^{r}_{s}a^{p}_{u} \\ - 2\gamma^{q}_{t}\gamma^{r}_{u}a^{p}_{s} - 2\gamma^{q}_{t}\gamma^{pr}_{su} + \gamma^{q}_{t}a^{pr}_{su} - 2\gamma^{q}_{u}\gamma^{r}_{s}a^{p}_{t} + 2\gamma^{q}_{u}\gamma^{r}_{t}a^{p}_{s} \\ + 2\gamma^{q}_{u}\gamma^{pr}_{st} - \gamma^{q}_{u}a^{pr}_{st} + \gamma^{r}_{s}\gamma^{pq}_{ut} + \gamma^{r}_{s}\gamma^{qp}_{tu} - \gamma^{r}_{s}a^{qp}_{tu} \\ + 2\gamma^{r}_{t}\gamma^{pq}_{su} - \gamma^{r}_{t}a^{pq}_{su} - 2\gamma^{r}_{u}\gamma^{pq}_{st} + \gamma^{r}_{u}a^{pq}_{st} + \gamma^{pq}_{st}a^{r}_{u} \\ - \gamma^{pq}_{su}a^{r}_{t} - \gamma^{pq}_{ut}a^{r}_{s} - \gamma^{pr}_{st}a^{q}_{u} + \gamma^{pr}_{su}a^{q}_{t} - \gamma^{pr}_{tu}a^{q}_{s} \\ - \gamma^{qr}_{su}a^{p}_{t} + \gamma^{qr}_{tu}a^{p}_{s} - \gamma^{rq}_{st}a^{p}_{u} + \gamma^{pqr}_{stu} + \tilde{a}^{pqr}_{stu}\end{aligned} \tag{7}$$

$$a^{pqr}_{stu} = \tilde{a}^{pqr}_{stu} + \gamma^{p}_{s}a^{qr}_{tu} - \gamma^{p}_{t}a^{qr}_{su} - \gamma^{p}_{u}a^{qr}_{ts} - \gamma^{q}_{s}a^{pr}_{tu} + \gamma^{q}_{t}a^{pr}_{su} - \gamma^{q}_{u}a^{pr}_{st} - \gamma^{r}_{s}a^{qp}_{tu} - \gamma^{r}_{t}a^{pq}_{su} + \gamma^{r}_{u}a^{pq}_{st}$$

$$
\begin{aligned}
&+\gamma^{pq}_{st}a^r_u-\gamma^{pq}_{su}a^r_t-\gamma^{pq}_{ut}a^r_s-\gamma^{pr}_{st}a^q_u+\gamma^{pr}_{su}a^q_t-\gamma^{pr}_{tu}a^q_s-\gamma^{qr}_{su}a^p_t+\gamma^{qr}_{tu}a^p_s-\gamma^{rq}_{st}a^p_u\\
&-2\gamma^p_s\gamma^q_ta^r_u-2\gamma^p_s\gamma^r_ua^q_t-2\gamma^p_t\gamma^q_ua^r_s-2\gamma^p_t\gamma^r_sa^q_u-2\gamma^p_u\gamma^q_sa^r_t-2\gamma^p_u\gamma^r_ta^q_s-2\gamma^q_s\gamma^r_ta^p_u-2\gamma^q_t\gamma^r_ua^p_s-2\gamma^q_u\gamma^r_sa^p_t\\
&2\gamma^p_s\gamma^q_ua^r_t+2\gamma^p_s\gamma^r_ta^q_u+2\gamma^p_t\gamma^q_sa^r_u+2\gamma^p_t\gamma^r_ua^q_s+2\gamma^p_u\gamma^q_ta^r_s+2\gamma^p_u\gamma^r_sa^q_t+2\gamma^q_s\gamma^r_ua^p_t+2\gamma^q_t\gamma^r_sa^p_u+2\gamma^q_u\gamma^r_ta^p_s\\
&-\gamma^p_s\gamma^{qr}_{tu}+\gamma^p_t\gamma^{qr}_{su}+\gamma^p_u\gamma^{rq}_{st}+\gamma^q_s\gamma^{pr}_{tu}-\gamma^q_t\gamma^{pr}_{su}+\gamma^q_u\gamma^{pr}_{st}+\gamma^r_s\gamma^{pq}_{ut}+\gamma^r_t\gamma^{pq}_{su}-\gamma^r_u\gamma^{pq}_{st}\\
&+4\gamma^p_s\gamma^q_t\gamma^r_u-4\gamma^p_s\gamma^q_u\gamma^r_t-4\gamma^p_t\gamma^q_s\gamma^r_u+4\gamma^p_t\gamma^q_u\gamma^r_s+4\gamma^p_u\gamma^q_s\gamma^r_t-4\gamma^p_u\gamma^q_t\gamma^r_s+\lambda^{pqr}_{stu}
\end{aligned}
\tag{8}
$$

5 Spin Free Result

$$
\begin{aligned}
E^{pqr}_{stu}\approx{}&D^p_sE^{qr}_{tu}-\frac{D^p_tE^{qr}_{su}}{2}-\frac{D^p_uE^{qr}_{ts}}{2}-\frac{D^q_sE^{pr}_{tu}}{2}+D^q_tE^{pr}_{su}-\frac{D^q_uE^{pr}_{st}}{2}-\frac{D^r_sE^{qp}_{tu}}{2}-\frac{D^r_tE^{pq}_{su}}{2}+D^r_uE^{pq}_{st}\\
&+D^{pq}_{st}E^r_u-\frac{D^{pq}_{su}E^r_t}{2}-\frac{D^{pq}_{ut}E^r_s}{2}-\frac{D^{pr}_{st}E^q_u}{2}+D^{pr}_{su}E^q_t-\frac{D^{pr}_{tu}E^q_s}{2}-\frac{D^{qr}_{su}E^p_t}{2}+D^{qr}_{tu}E^p_s-\frac{D^{rq}_{st}E^p_u}{2}\\
&-2D^p_sD^q_tE^r_u-2D^p_sD^r_uE^q_t-\frac{D^p_tD^q_uE^r_s}{2}-\frac{D^p_tD^r_sE^q_u}{2}-\frac{D^p_uD^q_sE^r_t}{2}-\frac{D^p_uD^r_tE^q_s}{2}-\frac{D^q_sD^r_tE^p_u}{2}-2D^q_tD^r_uE^p_s-\frac{D^q_uD^r_sE^p_t}{2}\\
&+D^p_sD^q_uE^r_t+D^p_sD^r_tE^q_u+D^p_tD^q_sE^r_u+D^p_tD^r_uE^q_s+D^p_uD^q_tE^r_s+D^p_uD^r_sE^q_t+D^q_sD^r_uE^p_t+D^q_tD^r_sE^p_u+D^q_uD^r_tE^p_s\\
&-D^p_sD^{qr}_{tu}+\frac{D^p_tD^{qr}_{su}}{2}+\frac{D^p_uD^{rq}_{st}}{2}+\frac{D^q_sD^{pr}_{tu}}{2}-D^q_tD^{pr}_{su}+\frac{D^q_uD^{pr}_{st}}{2}+\frac{D^r_sD^{pq}_{ut}}{2}+\frac{D^r_tD^{pq}_{su}}{2}-D^r_uD^{pq}_{st}\\
&+4D^p_sD^q_tD^r_u-2D^p_sD^q_uD^r_t-2D^p_tD^q_sD^r_u+D^p_tD^q_uD^r_s+D^p_uD^q_sD^r_t-2D^p_uD^q_tD^r_s
\end{aligned}
\tag{9}
$$